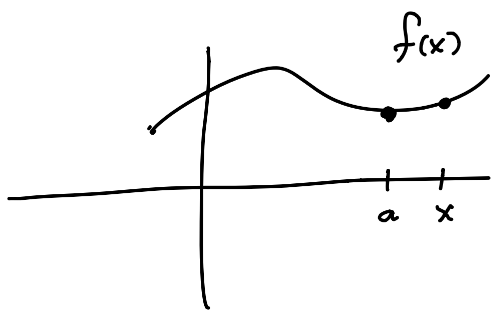
Section 2.8 Derivative as a Function

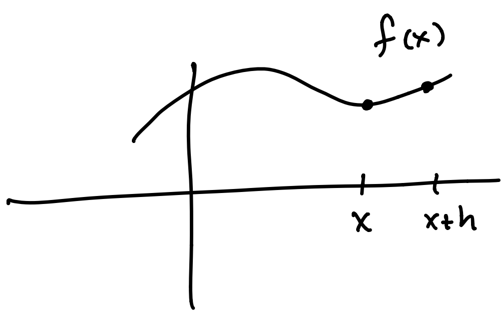


f' is notation for the derivative

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

This will give us a number for the derivative where $x = a$.

Another way to compute this...



$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This will give us a function for the derivative.

Ex ① $f(x) = x^2$ find $f'(4)$ both ways.

A) $\displaystyle\lim_{x \to 4} \frac{f(x) - f(4)}{x - 4}$

$$= \lim_{x \to 4} \frac{x^2 - 4^2}{x - 4}$$

$$= \lim_{x \to 4} \frac{(x-4)(x+4)}{x - 4}$$

$$= \lim_{x \to 4} (x + 4) = 4 + 4 = 8$$

B) $$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h}$$

$$= \lim_{h \to 0} 2x + h = 2x$$

Now use $x = 4$, $f'(4) = 2(4) = 8$

Ex ② $f(x) = 5x^2 + 3x + 1$

Where is $f'(x) = 0$?

You can use either definition to find the derivative.

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$ ← use this one if you like factoring

$$= \lim_{x \to a} \frac{5x^2 + 3x + 1 - (5a^2 + 3a + 1)}{x - a}$$

$$= \lim_{x \to a} \frac{(5x^2 - 5a^2) + (3x - 3a)}{x - a}$$ factor by grouping

$$= \lim_{x \to a} \frac{5(x^2 - a^2) + 3(x - a)}{x - a}$$

$$= \lim_{x \to a} \frac{5\cancel{(x-a)}(x+a) + 3\cancel{(x-a)}}{\cancel{x-a}}$$

$$= \lim_{x \to a} 5(x+a) + 3$$

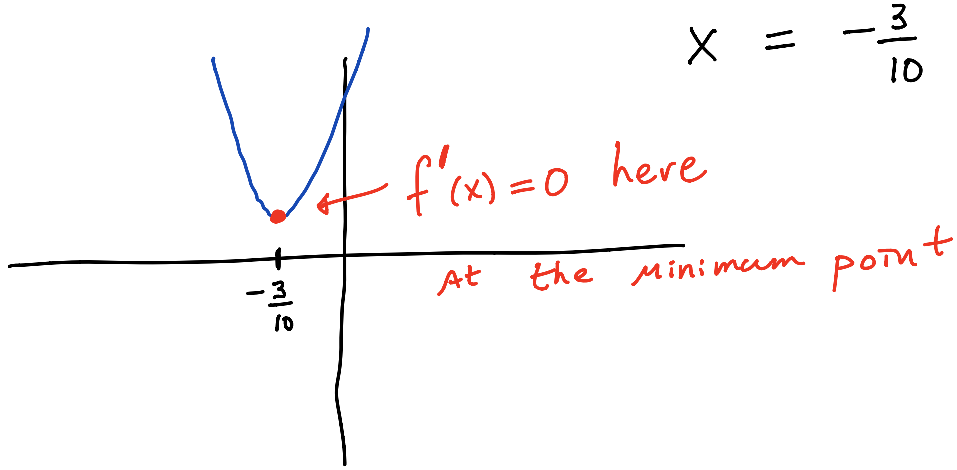
$$= \quad 5(a+a) + 3$$

$= 10a + 3$ is the derivative

or $10x + 3$ is also the derivative

Where does $10x + 3 = 0$?

$$x = -\frac{3}{10}$$



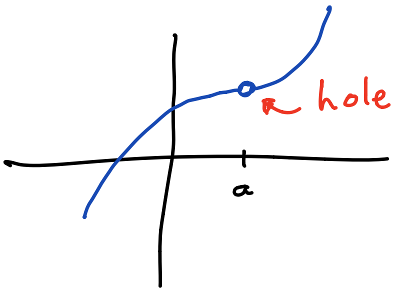
Notations for the derivative

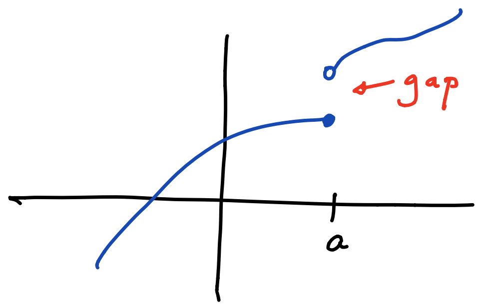
y' $\qquad f'(x)$ $\qquad \frac{dy}{dx}$ $\qquad \frac{d(\ )}{dx}$

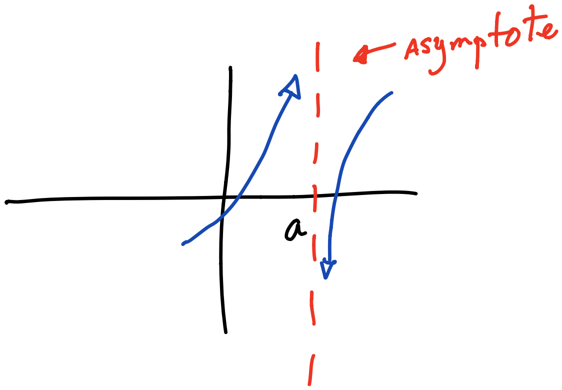
Does the derivative Always exist?

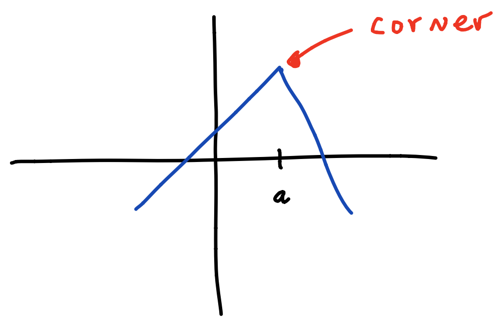
No, the function must be continuous And cannot have any cusps or corners.

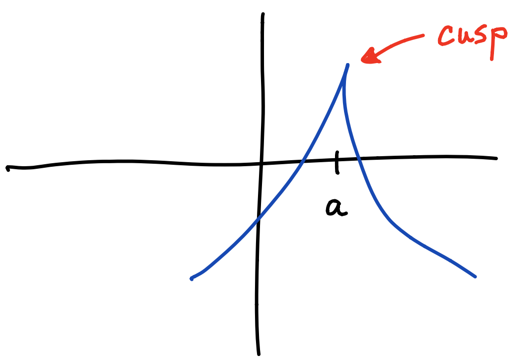
Here are example where the derivative does **NOT** exist at $x = a$.

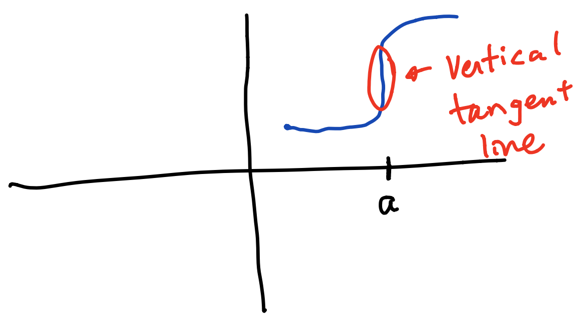






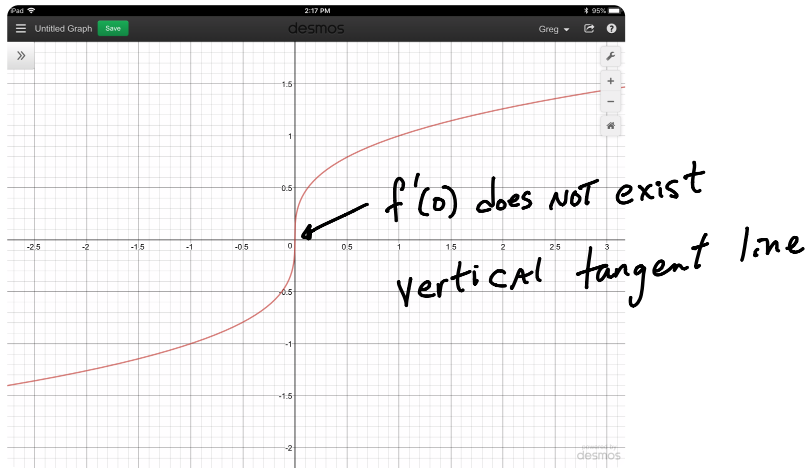




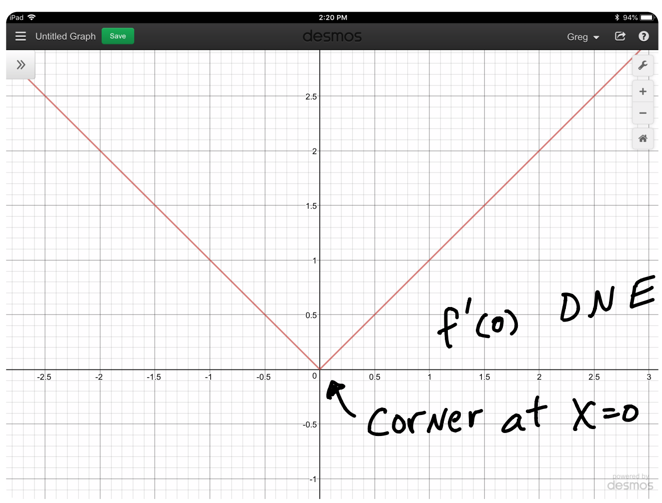


Ex ③ Graph the functions and show where the derivative does NOT exist.

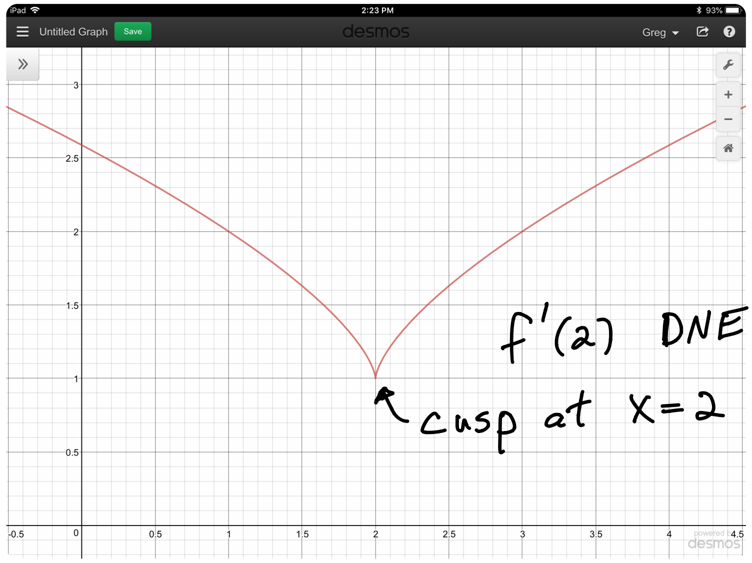
A) $f(x) = \sqrt[3]{x}$



B) $f(x) = |x|$



c) $f(x) = (x-2)^{2/3} + 1$



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$f'(2)$ DNE
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